

PROFESSIONAL SUMMARY

Chemistry undergraduate student and self-directed developer actively learning and practicing full-stack web development through real projects. Hands-on experience developing frontend user interfaces, backend APIs, relational databases, authentication mechanisms, role-based access control, admin portals, e-commerce workflows, and AI integration. Employs AI-assisted development tools including Antigravity IDE to accelerate prototyping and implementation while diligently testing, analyzing, and refining code quality.

TECHNICAL SKILLS

LANGUAGES

C • C++ • Python • JavaScript • TypeScript

FRONTEND

React • Vite • Tailwind CSS • Bootstrap • HTML • CSS

BACKEND & APIS

Python • FastAPI • REST APIs

DATABASES

PostgreSQL • Supabase • MySQL

CORE CONCEPTS

OOP • Data Structures • Algorithms • Auth & RBAC • API Integration

TOOLS & WORKFLOWS

Git • GitHub • Antigravity IDE • AI Development Tools

EDUCATION

Dhaka College

B.Sc. (Honours) in Chemistry

Current Status: 4th Year • Dhaka, Bangladesh

Rajshahi Cantonment Board School & College

Higher Secondary Certificate (HSC)

Passing Year: 2021 • GPA: 5.00 / 5.00

AI-ASSISTED WORKFLOW

Uses Antigravity IDE and AI-assisted development tools to accelerate exploration, prototyping, and debugging cycles. Approaches AI as a development aid, prioritizing critical review, code comprehension, logical testing, and manual refinement of generated code.

FEATURED PROJECTS

Dr. Chem — Interactive Chemistry Platform

EDUCATION

React • TypeScript • Tailwind CSS • Supabase / PostgreSQL • AI Integration

A Bangla-first chemistry learning platform designed for SSC and HSC students, featuring virtual chemistry labs with structured academic management tools.

- Engineered web application architecture with student learning paths, notes, question banks, and viva practice.
- Integrated AI tutoring functionality to help students understand complex chemistry concepts interactively.
- Implemented authentication and role-based workflows to separate student learning zones from teacher controls.
- Configured database-backed content management for lab resources, curriculum tracks, and assessments.

Corporate Technologies — E-Commerce Platform

E-COMMERCE

React • Vite • Tailwind CSS • Supabase • API Workflows

An e-commerce web platform for printing and office equipment, providing customer ordering flows and an operations admin portal.

- Developed core storefront: product catalog browsing, cart, wishlist, Buy Now, and order summary.
- Structured checkout workflows supporting delivery area selection, dynamic charges, and coupon processing.
- Configured multi-channel payment guides for Cash on Delivery (COD), bKash, Nagad, and Rocket instructions.
- Created admin dashboards for product management, customer oversight, and order status handling.

Business Workflow: Orders are placed, held for review, and confirmed via customer phone verification before dispatch processing.

Dhaka 24/7 Emergency Ambulance System

FULL-STACK

FastAPI • Python • SQLAlchemy • PostgreSQL • React • Vite • JWT • DaisyUI

An academic full-stack emergency dispatch system structured around distinct User, Driver, and Admin operations.

- Developed backend APIs using Python and FastAPI with SQLAlchemy ORM on PostgreSQL.
- Implemented JWT authentication alongside role-based authorization for Users, Drivers, and Admins.
- Built core booking flows incorporating pricing calculation, dispatch coordination, and fleet controls.

Personal Portfolio & Development Projects

WEB DEV

HTML • CSS • JavaScript • React • REST APIs • Git • GitHub

Personal developer portfolio and practice projects built to reinforce modern UI engineering, component lifecycle, and deployment workflows.

- Constructed responsive, accessible layouts with modular components and clear visual hierarchy.
- Strengthened REST API integration patterns and version control workflows using Git and GitHub.